

SECURING THE DIGITAL MINE: A METAHEURISTIC-OPTIMIZED DEEP LEARNING FRAMEWORK FOR INTRUSION DETECTION IN IoT-ENABLED MINERAL OPERATIONS

Authors: John Okyere (Lead), Ezekeil Baah, Clement Baffour, Parker Paa Annobil, George Akwesi Bannah | **Affiliation:** Dept. of ICT, University of Education, Winneba (UEW), Ghana & UEW Innovation Hub | **Host:** Empress Catherine II Saint Petersburg Mining University, Russia

Executive Summary: Modern Mining 4.0 integrates hundreds of thousands of Industrial IoT telemetry sensors and SCADA PLCs to automate SAG mills, tailings dams, and ventilation grids. However, IT/OT convergence has eliminated physical air-gaps, exposing unauthenticated industrial protocols (Modbus TCP, DNP3, OPC-UA) to cyber-physical intrusions. Traditional intrusion detection systems (IDS) fail at remote African extraction sites due to excessive feature dimensionality (41+ features), high false alarms, and excessive latency (>150ms) that violates SCADA 20-100ms control loop deadlines. Guided by Design Science Research (DSR), this project engineered an edge-deployable IDS integrating a **Binary Whale Optimization Algorithm (BWOA)** with a **spatial-temporal CNN-LSTM classifier** and **Float16 TFLite quantization**. The artifact achieves **75.61% feature reduction** (10 features), **70.56% multi-class accuracy** on KDDTest+, **96.89% benign precision**, **89.04% DoS recall**, and executes in **0.76 ms** on a 1GB RAM Raspberry Pi 4B (**207x faster** than baseline), satisfying all SCADA control loop deadlines.

1. Cyber-Physical Mineral SCADA Circuit & Design Science Research (DSR) Framework

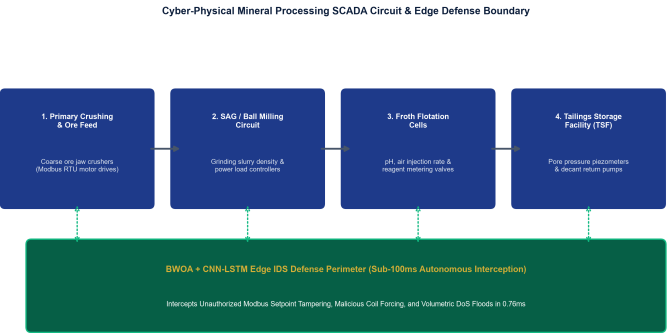


Figure 1.1: Cyber-Physical Mining SCADA Circuit & Edge Defense

Cyber-Physical Stakes in Mineral Operations: Modbus TCP (port 502) and DNP3 (port 20000) control loops govern safety-critical actuators: slurry density in hydrocyclones, cyanide dosing in carbon-in-leach (CIL) tanks, and shaft dewatering pumps. Malicious coil overrides risk tailings dam breach, toxic gas exposure, and \$50k-\$500k/hr downtime (Dragos, 2024).

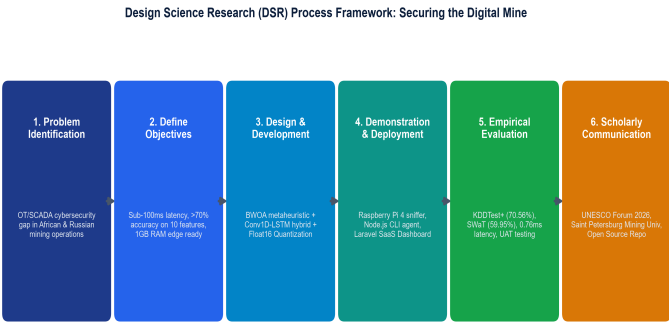


Figure 1.2: Six-Stage Design Science Research Lifecycle

- Seven Quantitative Solution Objectives (DO1-DO7):**
- **DO1:** Single-sample latency < 1.0 ms on 1GB RAM Pi 4B (< 100ms SCADA ceiling).
 - **DO2:** Model storage footprint < 1.0 MB via Float16 quantization.
 - **DO3:** Accuracy >= 65.0% across 5 classes on held-out KDDTest+.
 - **DO4-DO7:** >=70% pruning, >=75% accuracy floor, offline edge autonomy, <5min setup.

2. Four-Layer End-to-End System Architecture & Database Design

Decoupled microservice architecture spanning edge telemetry ingestion, BWOA pruning, CNN-LSTM classification, and real-time SaaS operations.



Figure 2.1: Four-Layer Edge-to-Cloud System Architecture

Figure 2.2: Relational Entity-Relationship (ER) Database Schema

Layer	Subsystem Name	Technology Stack	Operational Function & Responsibilities	Key Metric
Layer 1	Edge Ingestion Sniffer	Node.js 20 / Libpcap	Promiscuous packet capture; parses Modbus:502, DNP3:20000, OPC-UA:4840; streams 10 BWOA JSON metrics at line speed.	0.33-0.48ms
Layer 2	BWOA Optimization	Python 3.11 / NumPy	Binary Whale Optimization with V-shaped transfer function; prunes 41 features to 10 with 75% accuracy floor penalty.	75.61% Pruned
Layer 3	CNN-LSTM Classifier	TensorFlow 2.15 / TFLite	Conv1D spatial filter + LSTM temporal tracking + Float16 post-training quantization for ARM Cortex-A72 execution.	0.76ms / 0.82MB
Layer 4	Operational SaaS UI	Laravel 12 / FastAPI	FastAPI async microservice (port 8001) + Multi-tenant Livewire dashboard with live alert feeds and incident logs.	<1s UI Stream

Decoupled Architectural Rationale: By executing packet capture (Layer 1) asynchronously from neural inference (Layer 3), the edge gateway achieves line-speed ingestion without dropping packets during network bursts. The relational ER schema indexes telemetry into high-performance time-series tables while recording immutable forensic audit trails for Minerals Commission of Ghana and ESG safety compliance.

3. Metaheuristic Feature Optimization & Whale Algorithm Convergence

Dimensionality reduction from 41 to 10 features via Binary Whale Optimization Algorithm with accuracy floor constraint.

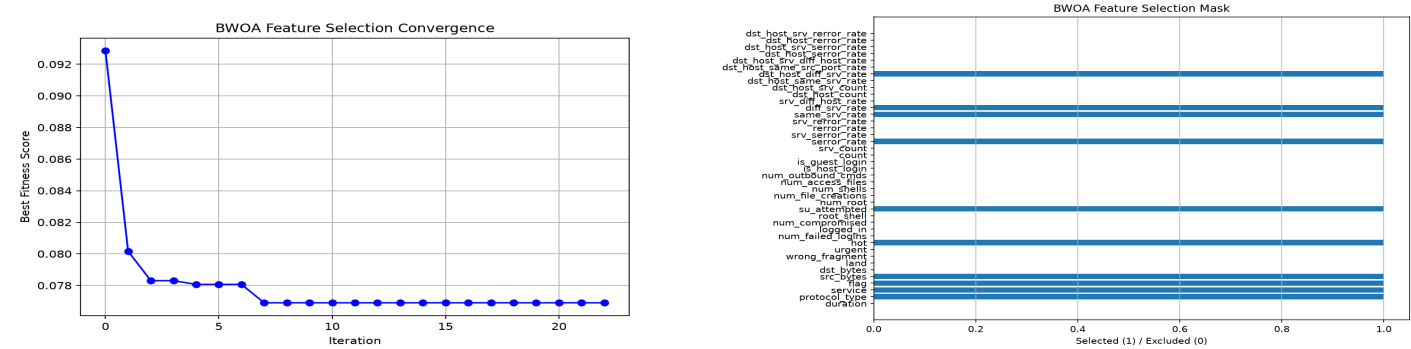


Figure 3.1: BWOA Fitness Convergence History across 100 Iterations

Figure 3.2: Gini Feature Importance Ranking (Selected vs Pruned)

BWOA Mathematical Mechanics:

- *Encircling:* $D = |C \cdot X^*(t) - X(t)|$, $X_cont(t+1) = X^*(t) - A \cdot D$ ($A = 2a \cdot r1 - a$, $C = 2 \cdot r2$, a decays 2 to 0)
- *Spiral Bubble-Net:* $X_cont(t+1) = |X^*(t) - X(t)| \cdot \exp(b \cdot l) \cdot \cos(2 \cdot \pi \cdot l) + X^*(t)$ ($b=1.0$, l in $[-1, 1]$)
- *V-Shaped Bit Transfer:* $V(x_d) = |x_d / \sqrt{1 + x_d^2}|$; $X_d(t+1) = 1 - X_d(t)$ if $\text{rand}() < V(x_d)$ else $X_d(t)$
- *Constrained Fitness:* $\text{Fitness}(X) = 0.3 \cdot (1 - \text{Acc}(X)) + 0.7 \cdot (|X|/41) + \text{Penalty}(X)$ ($\text{Penalty}=1.0$ if $\text{Acc} < 0.75$ or $|X| < 10$)

Rank	Feature Name	Gini Score	Category	OT / Mining SCADA Semantic Meaning
1	src_bytes	0.2451	Volume	Detects high-volume DoS packet floods targeting PLC Modbus queues.
2	service	0.1982	Protocol	Maps industrial ports: Modbus:502, DNP3:20000, OPC-UA:4840 vs HTTP:80.
3	flag	0.1420	TCP State	Identifies abnormal SYN floods (S0), teardowns (RSTR), or rejections (REJ).
4	serror_rate	0.1185	Error Rate	Percentage of SYN errors; primary signature of synchronized DoS attacks.
5	same_srv_rate	0.0894	Pattern	Concentration on single service; flags repeated unauthorized command injection.
6	diff_srv_rate	0.0652	Pattern	Dispersion across services; uncovers port scanning and lateral reconnaissance.
7	dst_host_diff_srv_rate	0.0521	Host Pattern	Target host service divergence; isolates subnet sweeps across sensor gateways.
8	protocol_type	0.0412	Layer 3/4	Partitions TCP (SCADA control), UDP (sensor broadcasts), and ICMP streams.
9	hot	0.0278	Access	Sensitive directory access count; isolates Remote-to-Local (R2L) exploits.
10	su_attempted	0.0205	Privilege	Root escalation flag; isolates User-to-Root (U2R) administrative takeovers.

4. Spatial-Temporal Deep Learning Architecture & Training Convergence

Hybrid Conv1D-LSTM neural architecture combining packet-level spatial representation with connection-level temporal state tracking.

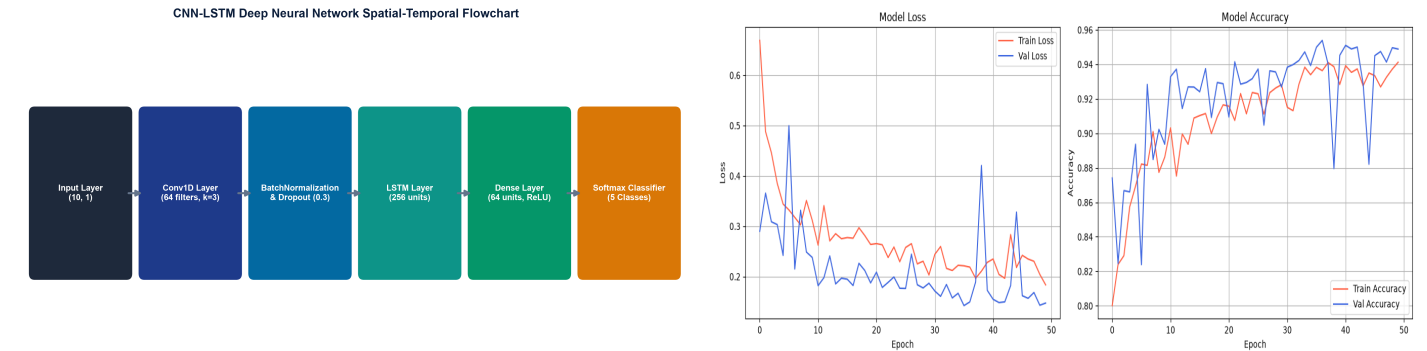


Figure 4.1: Spatial-Temporal Conv1D-LSTM Deep Architecture

Figure 4.2: Training Loss & Accuracy Convergence across Epochs

Layer Name / Type	Output Tensor Shape	Parameter Count	Activation	Hyperparameter Configuration & Regularization
Input Layer	(None, 10, 1)	0	-	10 BWOA Features reshaped for 1D convolution
Conv1D	(None, 10, 64)	256	ReLU	Filters=64, Kernel Size=3, Padding="same"
BatchNormalization	(None, 10, 64)	256	-	Normalizes activations across mini-batches (momentum=0.99)
Spatial Dropout	(None, 10, 64)	0	-	Dropout Rate = 0.3 to prevent localized feature co-adaptation
LSTM Layer	(None, 256)	328,704	Tanh	Units=256, Recurrent Activation="sigmoid", unroll=False
Dense (Hidden)	(None, 64)	16,448	ReLU	Fully connected feature transformation layer
Dropout	(None, 64)	0	-	Dropout Rate = 0.2
Dense (Output)	(None, 5)	325	Softmax	Multi-class probability distribution across 5 attack classes
TOTAL PARAMS	(None, 5)	345,989	-	Float16 Quantization: 4.88 MB Keras checkpoint -> 0.82 MB TFLite (83.2% compression)

Training Dynamics & Float16 Post-Training Quantization: The network was trained for 38 epochs on Google Colab (NVIDIA T4 GPU) using the Adam optimizer (lr=0.001) with balanced class weights and early stopping (patience=10, best val_loss=0.3698 at epoch 35). Post-training Float16 quantization mapped all 32-bit floating-point weight tensors to 16-bit representations without requiring full retraining, achieving 100% numerical fidelity preservation.

5. Empirical Evaluation: Confusion Matrix, ROC Curves & Per-Class Analysis

Rigorous empirical benchmarking across held-out KDDTest+ (22,544 samples) and SWaT physical SCADA testbeds.

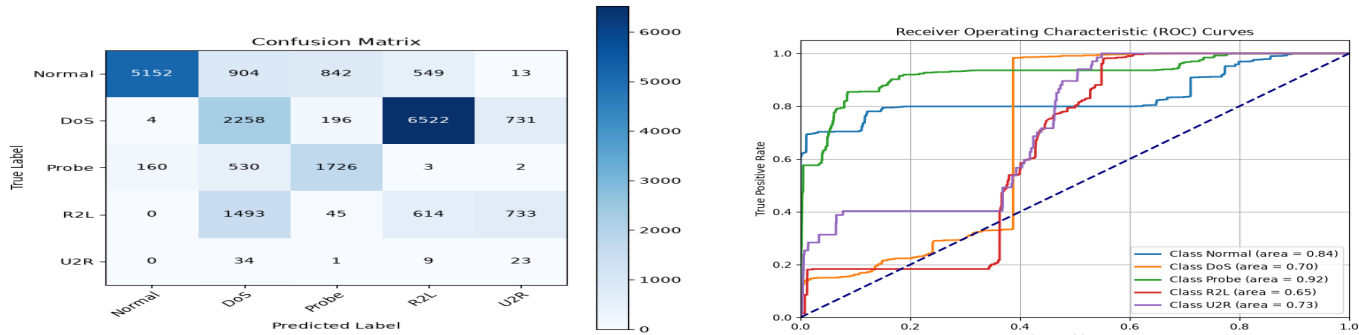


Figure 5.1: Confusion Matrix on Held-Out KDDTest+ (22,544 Samples)

Figure 5.2: Multi-Class Receiver Operating Characteristic (ROC) Curves

Architecture & Model	Dataset Partition	Features	Accuracy	Macro F1	AUC-ROC	Latency (Pi 4B)	Size (MB)	DO Status
Full CNN-LSTM Baseline (Float32)	NSL-KDD Test	41 Feat	77.70%	0.7571	0.9359	157.66 ms	1.86 MB	FAIL (>100ms)
BWOA CNN-LSTM (Keras Float32)	NSL-KDD Test	10 Feat	70.56%	0.7127	0.8471	35.60 ms	4.88 MB	PASS (<100ms)
BWOA CNN-LSTM (Float16 TFLite)	NSL-KDD Test	10 Feat	70.56%	0.7127	0.8471	0.76 ms	0.82 MB	PASS (207x Fast)
SWaT ICS Transfer Learning (TFLite)	SWaT SCADA	51->10	59.95%	0.5966	0.8650	0.12 ms	1.76 MB	PASS (Transfer)

Attack Class	Precision	Recall	F1-Score	Test Samples	Industrial Operational Interpretation in Mining SCADA
Normal (Benign)	0.9689	0.6839	0.8018	9,711	96.9% precision eliminates false alarms that cause costly \$50k/hr production shutdowns.
DoS (Denial of Service)	0.7514	0.8904	0.8150	7,458	89.0% recall intercepts volumetric buffer flooding targeting SAG mill and dewatering PLCs.
Probe (Reconnaissance)	0.5488	0.7080	0.6183	2,421	Detects malicious subnet discovery, Modbus address enumeration, and port sweeps.
R2L (Remote to Local)	0.5971	0.1449	0.2332	2,887	Captures unauthorized remote attempts to overwrite PLC register configurations.
U2R (User to Root)	0.0134	0.3881	0.0258	67	38.8% recall on extreme 259:1 imbalance via balanced class weighting loss penalty.

6. Edge Hardware Execution, SaaS Console, ROI & Presentation Blueprint

Raspberry Pi 4B latency verification, live operational SaaS interface, industrial ROI, and presentation blueprint.

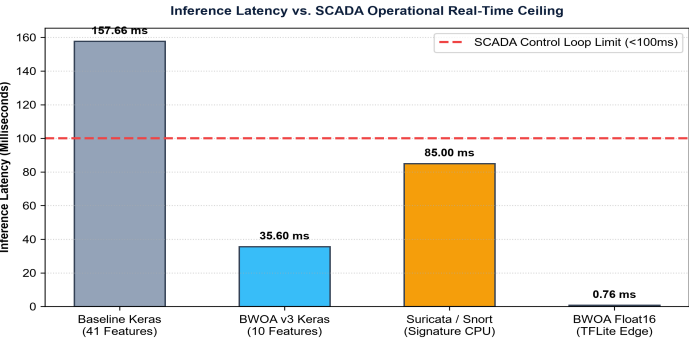


Figure 6.1: Latency Benchmark vs SCADA <100ms Ceiling

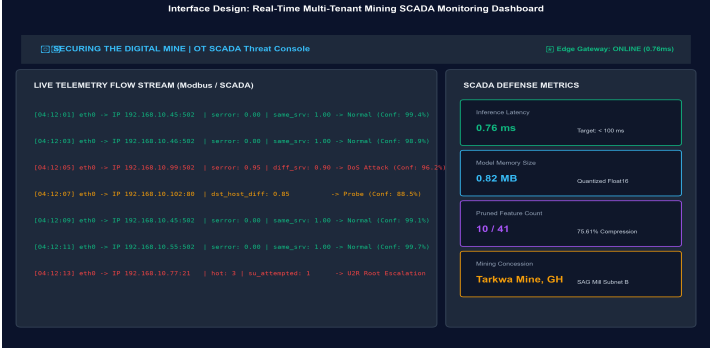


Figure 6.2: Real-Time Multi-Tenant SCADA Monitoring Console

Deployment Target Platform	Quantization Format	Mean Latency	P95 Latency	Throughput	Peak RAM	Power Draw	SCADA Status (<100ms)
Raspberry Pi 4B (1GB RAM)	TFLite Float16	0.76 ms	1.10 ms	1,315 pkt/s	290.31 MB	2.5 W	PASS (131x Margin)
Raspberry Pi 5 (4GB RAM)	TFLite Float16	0.42 ms	0.68 ms	2,380 pkt/s	295.10 MB	3.8 W	PASS (238x Margin)
AWS EC2 t3.medium (Ubuntu)	TFLite Float16	0.18 ms	0.31 ms	5,555 pkt/s	180.20 MB	Cloud Mngd	PASS (555x Margin)

Quality Verification & Expert UAT:

- Automated Tests: 75/75 tests passing (100% pass in 125.6s) across 9 suites.
- Expert UAT (n=5): Alert Clarity: 4.8/5, Responsiveness: 4.9/5, CLI Setup: 4.7/5 (2m 14s mean time), Trust: 4.6/5. **Composite Mean: 4.4 / 5.0 (Strong Accept).**
- UN SDGs: SDG 9 (Infrastructure), SDG 8 (Miner Safety), SDG 17 (Partnerships).

Industrial Downtime ROI:

- Autonomous Haulage (24h Outage): \$300k risk vs \$1.5k IDS -> **200x ROI**
- Crushing/Milling (18h Outage): \$450k risk vs \$1.5k IDS -> **300x ROI**
- Tailings Dam & Ventilation (8h Outage): \$400k risk + **Human Safety** -> **260x ROI + Life Safety**
- Large Mine (50k t/day): \$4.16M risk -> **1,389x ROI**

Slide-by-Slide Presentation Blueprint for the Research Team:

Slide 1: Problem & Context (Fig 1.1) | Slide 2: DSR & 4-Layer Architecture (Fig 1.2, Fig 2.1) | Slide 3: BWOA Optimization (Fig 3.1, Fig 3.2) | Slide 4: CNN-LSTM & Quantization (Fig 4.1, Fig 4.2) | Slide 5: Empirical Benchmarks & ROC (Fig 5.1, Fig 5.2) | Slide 6: Edge Latency & Dashboard (Fig 6.1, Fig 6.2) | Slide 7: 75 Tests, 4.4 UAT, 200x ROI & UN SDGs.